MIL-C-40137E
20 March 1984
SUPERSEDING
MIL-C-40137D
21 February 1978

MILITARY SPECIFICATION

CLEVISES, RIGGING, AIR DELIVERY TYPE: GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the general requirements for three types of air delivery rigging clevises, commonly referred to as shackles. Specific design requirements for a particular clevis are covered by the applicable MS Sheet Form Standard.

1.2 Classification. Clevises shall be of the following types and part numbers as specified (see 6.2):

Type	Description	Part Number
I	Clevis, suspension	MS 70087-1
I	Clevis, suspension	MS 70087-2
I	Clevis, suspension	MS 70087-3
I	Clevis, suspension	MS 70087-4
I	Clevis, suspension	MS 70087-5
II	Clevis, tie-down	MS 70085-1
II	Clevis, tie-down	MS 70085-2
III	Clevis, riser	MS 70086-1

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MS 70085 - Clevises, Tie-down, Air Delivery, Type II
MS 70086 Clevises, Riser, Air Delivery, Type III
MS 70087 - Clevises, Suspension, Air Delivery, Type I

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order or precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.4) and components. Materials and components used shall conform to the documents and standards referenced on the applicable MS standard.

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3.3 Construction. Fabrication of components and assembly of clevises shall conform to the requirements of this document and the design details shown or specified on MS 70085, MS 70086, and MS 70087, as applicable.

3.3.1 Cross section dimensions. Cross sections of forged components may be cut out of round or unsymmetrical due to die mismatch, flash trim and cleaning processes. The cross section contour at any point shall be within the maximum and minimum material condition limits set by the dimensions and tolerances specified.

3.3.2 Heat treatment. Forged clevis components shall be heat treated as specified on the applicable MS standard.

3.4 Finish.

3.4.1 Surface defects and roughness. Forged components shall be free from scale, burrs, sharp edges, fins, cracks, and laps. Forging flash shall be removed substantially flush with the adjacent surfaces. Surface roughness shall not exceed values specified on the applicable MS standard. Any suitable tactual microfinish comparator may be used for evaluation.

3.4.2 Protective. Protective finish for metal components, excluding AN and MS standard hardware, shall be as specified on the applicable MS standard.

3.5 Performance. Clevises shall be capable of withstanding the proof loads specified on the applicable MS standard without developing any evidence of failure, visible permanent deformation, or defects that would interfere with disassembly when tested in 4.4.4.

3.5.1 Functional performance. The hardware shall meet all functional test and assembly requirements specified on the applicable military standards.

3.6 Marking for indentification. Each clevis body shall be marked for identification with the part number and manufacturer's name or readily identifiable trademark. Forged clevis pins shall be marked with the part number only. The marking shall be the size and in the location shown on the applicable MS standard. The manufacturer may, at his option, utilize either raised or indented marking. If raised marking is elected, the height of characters shall be sufficient to insure legibility but shall not exceed 0.02 inches.

3.7 Workmanship. All components and assemblies of the rigging clevises shall be free of any rust or corrosion, dirt and other foreign matter. External surfaces shall be free from sharp edges or burrs. Components shall not be fractured, cracked, buckled, loose, dented, missing, malformed or misaligned.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested for proof load as specified in 4.5.1. The presence of any defect or failure of the proof load test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of clevises. The sample unit shall be one clevis. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating to metal	X	
	Any area of rust or corrosion	X	
	Dirt, oil, grease, or other foreign matter		X
Design	Not specified type or size	X	
	Any departure from design indicated on applicable MS standard (not otherwise classified herein)	X	
Construction and workmanship	Any component missing	X	
	Any component cracked or fractured	X	
	Any operation omitted or not performed as specified	X	
	Any sharp edges, burrs, or metal slivers	X	
	Any part not assembled or joined as specified		X
	Any surface roughness exceeding specified limit		X
Identification marking	Missing, illegible, incomplete, incorrect, not permanent		X

4.4.3 End item critical dimension examination. The end item shall be examined for conformance to the critical dimensions specified on the applicable military standard. The lot size shall be expressed in units of clevises of one type and part number only. The sample shall be the number of clevises indicated by inspection level S-4 except that no sample shall be smaller than ten clevises regardless of lot size. Failure of any clevis to conform to any critical dimension shall be cause for rejection of the lot.

4.4.4 End item testing. The end item shall be proofloaded as specified in 4.5.1 and examined for conformance to the requirements in 3.5. The lot size shall be expressed in units of clevises of one type and part number only. The sample shall be the number of clevises indicated by inspection level S-2. Any failure to conform to the requirements in 3.5 shall be cause for rejection of the lot.

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4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number of clevises per container is more or less than specified

4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of inmproper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Proofload test. The proofload specified on the applicable Military Standard (MS) shall be applied for not less than 15 seconds in the direction shown on the standard. Any suitable tensile testing machine may be used. The attachments to the clevis for proofloading shall be made as follows:

a. Attachment at pin end. Use a steel band of appropriate thickness and a width approximately equal to the clevis mortise dimension.

b. Attachment at bow end.

1. For bow with a straight section at the top, use a steel band of approximate thickness and a width approximately equal to the length of the straight section.

2. For bow with a continuous curvature use a steel band or bar of a width or diameter not greater than the clevis pin.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Clevises of one description only shall be unit packed in the most compact manner in quantities as specified in table II within a snug-fitting fiberboard box conforming to style RSC, type CF, class WR, grade W5c of PPP-B-636. Each box shall be closed and waterproofed by means of tape in accordance with the appendix of PPP-B-636.

TABLE II. Quantity of clevises per unit pack and shipping container

Type	Part number	Quantity per unit pack (each)	Quantity shipping container (each)
I	MS 70087-1	10	60
I	MS 70087-2	12	48
I	MS 70087-3	4	16
I	MS 70087-4	4	16
I	MS 70087-5	4	16
II	MS 70085-1	48	192
II	MS 70085-2	10	60
III	MS 70086-1	60	360

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5.1.2 Commercial. Clevises shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Clevises of one description only, preserved as specified in 5.1, shall be packed in the quantities as specified in table II within a snug-fitting shipping container conforming to overseas type, style A or J, grade A or B type 2 load of PPP-B-601, as specified (see 6.2). Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Clevises of one description only, preserved as specified in 5.1, shall be packed in the quantities as specifeid in table II within a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety DW) or SF, class domestic, grade 350 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard shipping container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with Method III as specified in the appendix of the container specification, except the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Clevises, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), clevises of one description only, packed as specified in 5.2.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the contaienr is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The clevises are intended for use in rigging cargo and equipment for air delivery.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Type and part number of clevis required (see 1.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing of clevises (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample and consist of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 11

Review activities:

Army - AV
Air Force - 82, 99

User activities:

Navy - MC, AS

Preparing activity:

Army - GL

Project No. 1670-0686

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-40137E	Clevises, Rigging, Air Delivery Type: General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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